

From TBMP to TSMP: Juneau's Tourism Never Ends

With the development of tourism in Juneau, tourists are pouring into the city in an endless stream. While creating huge wealth, the living space of local residents has been squeezed. At the same time, the Mendenhall Glacier is gradually retreating. To resolve these contradictions, we develop a **sustainable tourism measure plan (TSMP)**, aiming to balance tourism development with the well-being of local residents and environmental protection.

Several models are established: Model I: Multi-Objective Optimization Model for Sustainable Tourism; Model II: Expenditure-Feedback Model for Additional Revenue; Model III: Expansion Model for the Maldives.

For Model I: We built a multi-objective optimization model, aiming to maximize fiscal revenue, minimize environmental damage, and maximize residents' satisfaction. **Regression analysis, the analytic hierarchy process (AHP)** were used to construct the objective functions. The **NSGA-II Algorithm** was applied to search for the Pareto-optimal solution set. Taking the data of 2023 as an example, the optimal measures for the next year was obtained: limit the tourist flow to **around 1.43 million**, increase the tourist fee by **72.4 USB per person**, and increase the hotel tax rate by **0.08**.

For Model II: This model refines Model I by expending the extra revenue from its optimal measures for circular benefits. By examining the decision variables in the optimal solution of Model I, we found that the variables related to **road area** and **infrastructure carrying capacity** are always near initial upper-bounds, highlighting their significance for overall efficiency. Therefore, we constructed an **expenditure allocation model** for these two indicators and a **feedback model** regarding residents' satisfaction was constructed.

Before promoting the model, we conducted a **sensitivity analysis** using the **local perturbation method** and presented the results in a **heatmap**. We measured the importance of decision variables by comparing correlation coefficients. The results show that limiting the number of tourists is the most important indicator. Increasing tourist fees is more important than hotel taxes. Additionally, tourist diversion also needs to be taken into account.

For Model III: To verify the generalizability of the model, we applied it to the **Maldives**, which also faces tourism sustainability issues. The model was adjusted to get the optimal solution, and a sensitivity analysis was conducted. The results showed both **consistency** with Juneau and **reasonable differences**, like a higher correlation between residents' satisfaction and tourist numbers. This fully shows the model's **universality**.

In addition, we also discussed the applicability of the model in low-tourist-volume areas and offered feasible plans for sustainable tourism development, based on previous findings.

Keywords: Sustainable Tourism; Multi-Objective Optimization; NSGA-II Algorithm; Sensitivity Analysis